

# Superstore Sales Analytics

An Excel project: data cleaning, analysis and an interactive dashboard

## Business problem

A US retailer of furniture, office supplies and technology wants to know where its growth comes from, which products and regions really make money, and where margin is being lost. This workbook cleans two years of order data, analyses it, and presents the answers in an interactive dashboard with written recommendations.

## Questions answered

1. How did sales, profit and margin change between the two years?
2. Which categories, sub-categories and regions make money, and which lose it?
3. Which states and products matter most?
4. How do returns, shipping mode and payment mode differ?
5. How loyal is the customer base and how concentrated are sales?
6. What should the business do next?

## Dataset

|          |                                                           |
|----------|-----------------------------------------------------------|
| Source   | Online Dataset                                            |
| Period   | 1 Jan 2019 to 31 Dec 2020                                 |
| Grain    | One row per order line (a product within an order).       |
| Size     | 5,901 order lines, 3,003 orders, 773 customers, 49 states |
| Currency | USD assumed.                                              |

## Workbook map

| Sheet                             | What it does                                                                            | Type          |
|-----------------------------------|-----------------------------------------------------------------------------------------|---------------|
| <a href="#">Cover</a>             | This page: overview, workbook map, assumptions and limits.                              | Presentation  |
| <a href="#">Dashboard</a>         | Interactive dashboard: four filters, 8 KPI cards, 7 charts and a top 10 products table. | Presentation  |
| <a href="#">Insights</a>          | Nine findings with live evidence and recommended actions.                               | Presentation  |
| <a href="#">Product Analysis</a>  | Year over year, category and sub-category profitability, returns and seasonality.       | Analysis      |
| <a href="#">Regional Analysis</a> | Regions, region by category heat maps, returns by region and a state scorecard.         | Analysis      |
| <a href="#">Customer Analysis</a> | Retention, sales concentration, top customers, segment, shipping and payment.           | Analysis      |
| <a href="#">Data Notes</a>        | Source, cleaning log, live validation checks and data dictionary.                       | Documentation |
| <a href="#">Data</a>              | Cleaned order lines (Excel table tblSales) plus formula helper columns.                 | Data          |
| <a href="#">Calc</a>              | Calculation engine that feeds the Dashboard: filter criteria and chart tables.          | Engine        |
| <a href="#">calc Products</a>     | Product list with sales. Feeds the top 10 products.                                     | Engine        |
| <a href="#">calc Customers</a>    | Customer list with sales and ranks. Feeds retention and top customers.                  | Engine        |

## How to use the dashboard

1. Open the Dashboard sheet and use the four yellow dropdowns (Year, Region, Segment, Category).
2. Every KPI card, chart and the top 10 products table updates. Set all four to All for the full picture.
3. Read the Insights sheet for the findings and recommended actions. Their evidence cells recalculate too.

## Excel techniques used

Named ranges, SUMIFS and COUNTIFS across the full dataset, INDEX with MATCH and LARGE for dynamic top-N lists, dropdown filters with data validation, native charts (combo chart with a secondary axis, doughnuts, clustered and diverging bars), conditional formatting (data bars and colour scales) and live validation checks. No macros and no pivot tables, so it opens and recalculates anywhere.

## Assumptions and limits

- There are no discount, cost or target columns, so the causes behind margin changes cannot be tested here.
- Return flags exist only for second-year orders, so return rates use that year only.
- Currency is assumed to be USD.
- 28 Product IDs carry more than one product name. Rankings use Product ID and show the first name found.
- Row ID 4918 appears on two unrelated orders. Row ID is not used as a key.
- A small number of lines show more profit than sales. They are kept and flagged (Product\_Analysis, section 7).